

Multiple Choice (10 marks)

1. Which foods tend to be consumed in lower quantities in Wales and Scotland (as of 2020)?
 - (a) Meat
 - (b) Grains
 - (c) Fruits and vegetables
 - (d) Legumes
 - (e) Seafood
2. Bronchial breathing is characterized by
 - (a) Inspiratory component louder and longer with a gap between inspiration and expiration
 - (b) Inspiratory component louder and longer, but no gap between expiration and inspiration
 - (c) Expiratory component louder and longer, but no gap between expiration and inspiration
 - (d) Both inspiratory and expiratory components are equal in length and volume
 - (e) Inspiratory component louder and longer with a gap between expiration and inspiration
3. How many chromosomes do all human somatic cells contain?
 - (a) 30
 - (b) 24
 - (c) 44
 - (d) 3
 - (e) 23
4. Tay-Sachs disease is caused by deficiency of _____?
 - (a) Alpha-L-iduronidase
 - (b) Arylsulfatase A
 - (c) Hexosaminidase B
 - (d) Glucose-6-phosphatase
 - (e) Lysosomal alpha-glucosidase

5. A difference between the social networks of older and younger adults is that older adults have
- (a) Larger networks overall
 - (b) Less frequent communication with their circles
 - (c) Smaller networks overall
 - (d) More dispersed circles
 - (e) Smaller inner circles

True / False (10 marks)

Answer True or False

- 6. True or False: Cohort effects can influence the prevalence of alcoholism across different age groups in society.
- 7. True or False: Identity is primarily determined by one's financial status and societal position within a community.
- 8. True or False: The corresponding intercostal nerve lies medial to the inferior border of the rib.
- 9. True or False: Young adults have the highest protein density requirements relative to their energy needs compared to other age groups.
- 10. True or False: The pituitary gland is situated in the dorsal body cavity, near the spinal cord.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the relationship between abnormal breathing patterns and head injuries, specifically analyzing the role of altered consciousness and how anaerobic respiration fits into this context.
- 12. Discuss the role of vitamin D as a lipid-soluble antioxidant in cell membranes, analyzing its significance in cellular health and potential implications for disease prevention.

End of Paper